

Rebecca Chou

Newton, Massachusetts | rstchou@gmail.com | (617) 487-2040 | rebeccaschou.com

Education

Brown University

August 2022 – May 2026

Sc.B. Computer Science & A.B. Literary Arts, Departmental Honors

GPA: 4.00/4.00

Relevant Coursework: Software Security Exploitation, Topics in Software Security, Applied Cryptography, Computer Networks, Introduction to Computer Systems Security, Introduction to Computer Systems

The Lawrenceville School

August 2018 – May 2022

High School Diploma, Valedictorian

GPA: 4.13/4.30

Research Experience

DirtyRand

Secure Systems Lab | Brown University

Kernel Exploitation

2025 – Present

- **Problem:** The Linux kernel random number generator is critical to system security, yet its internal state remains largely unprotected against memory corruption and data-only attacks.
- **Contribution:** Investigated vulnerabilities in Linux's RNG state, demonstrating how a restricted memory write can make downstream randomness predictable and enable attacks against ASLR, stack canaries, networking, and TLS. Proposed a hardware-assisted defense using Intel MPK and ARM MTE, which protects RNG state with minimal performance overhead.

Chirp

Human-Computer Interaction Lab | Brown University

Communication Systems

2023 – Present

- **Problem:** Text is often the dominant medium for online communication, but text-heavy systems can create barriers for people who cannot rely on written language.
- **Contribution:** Investigated emoji-only communication as an extreme case of low-text interaction through two complementary studies. Examined how users interpret and construct meaning from emoji-only messages. Analyzed longitudinal interaction data to examine how users sustain communication in a severely constrained setting. Findings show that users can sustain meaningful communication and collaboratively develop shared meanings, suggesting that effective low-text systems can support communication through social meaning-making rather than semantic precision alone.

BeeBox

Secure Systems Lab | Brown University

Kernel Protection

2024 – 2025

- **Problem:** Existing implementations of BeeBox, a mitigation for speculative-execution attacks against Linux BPF, support only a limited subset of BPF functionality, leaving its effectiveness and performance across broader use cases unclear.
- **Contribution:** Expanded and evaluated BeeBox against the Linux BPF kselftest suite, identifying and addressing incompatibilities across data structures, helper functions, and kernel execution paths. Extended BeeBox to support additional BPF functionality and documented requirements and issues in the kselftest suite, establishing the engineering and performance tradeoffs of hardening BPF in broader contexts.
- **Publication:** Anticipated extended journal submission.

Publications

Posters

- **Rebecca Chou**. “Unseed: Exploiting the Linux Workqueue to Break RNG Reseeding.” Presented in partial fulfillment of the requirements for Honors in the Department of Computer Science at Brown University, April 2026.
- **Rebecca Chou**, Anjali Upadhyaya, Danial Ahmad, Abhiraj Saxena, and Vakhtang Tchantchaleishvili. “Computational Sentiment Analysis of 2000–2020 AATS and STS Presidential Addresses.” November 2021.

Work Experience

Fidelity Investments

Boston, MA

Associate Software Engineer

August 2026 – Present

- Advisor: Cassandra Costello
- Design and deploy an enterprise trading platform to service real-time trade activity

Fidelity Investments

Boston, MA

Mobile Engineering Intern

June 2025 – August 2025

- Advisor: Manik Lamba
- Built the Trader+ saved orders experience, enabling active traders to save and manage orders in Fidelity’s flagship mobile app

Fidelity Investments

Boston, MA

Mobile Engineering Intern

June 2024 – August 2024

- Advisor: Matthew Berman
- Designed and implemented tooltip components and tutorial flows to aid feature discovery
- Migrated legacy API layers to modern architectures in the Fidelity flagship mobile app

Jefferson Health

Philadelphia, PA

Research Intern

June 2021 – November 2021

- Advisor: Vakhtang Tchantchaleishvili
- Performed machine learning sentiment analysis on two decades of American Association for Thoracic Surgery (AATS) and Society of Thoracic Surgeons (STS) presidential speeches

Teaching Experience

CSCI 0200: Data Structures & Algorithms

Brown University

Head Teaching Assistant

Fall 2024

- Led a 10-person teaching team to design homework and projects on data structures and algorithms, emphasizing real-world applications

CSCI 0200: Data Structures & Algorithms

Brown University

Teaching Assistant

Fall 2023, Spring 2024

- Collaborated with a 60+ person teaching team to teach 200+ students data structures and algorithms

MATH 0520: Linear Algebra

Brown University

Teaching Assistant

Spring 2023

- Led biweekly lectures and individual office hours to teach a 48-student section

Projects

NeoGrowth: Fenton 2025 & 2013 Newton, MA
iOS Mobile Application June 2025 – August 2025

- Developed a preterm growth tool for pediatricians based on newly released Fenton 2025 growth charts, with backwards support for Fenton 2013 growth charts
- Available on the Apple App Store (1.5k+ downloads)

PediTools: Fenton 2013 Newton, MA
iOS Mobile Application July 2018 – June 2019

- Developed a preterm growth tool for pediatricians based on the Fenton 2013 growth chart
- Previously available on the Apple App Store (13k+ downloads)

Service

University Service

• Technology Chair	Brown University Women in Computer Science (WiCS)	2024 – 2025
• Mentorship Chair	Brown University Women in Computer Science (WiCS)	2023 – 2024

Community Service

• Fundraising Volunteer	The Tomorrow Fund Childhood Cancer Foundation	2023 – 2026
• Mobile App Developer	Roc Solid Foundation Childhood Cancer Foundation	2023 – 2024

Awards

• Sigma Xi Nominated Member	Brown University	2026
• Senior Prize in Computer Science	Brown University	2026
• Departmental Honors in Computer Science	Brown University	2026
• Undergraduate Teaching & Research Award	Brown University	2025
• Valedictorian	The Lawrenceville School	2022
• Cum Laude Society	The Lawrenceville School	2022

Other Interests

Taekwondo: Six-time national collegiate medalist and former instructor for Brown Taekwondo
Creative Writing: Creative nonfiction and poetry with a digital and cross-disciplinary focus
Crochet: Amigurumi birds, curtains, cardigans, and anything else that brings joy!